

IXplore IX73 / IX83

Inverted Imaging Platform

Each system in the IXplore inverted microscope series is tailored to fit a specific research application to help scientists more efficiently accomplish their goals. IXplore systems provide accurate, reproducible images and data, and can be adapted as experimental needs evolve or become increasingly complex over time.

IX73



IX83



IXplore Standard

Compound Microscope System

Designed for simple multicolor fluorescence imaging and routine experiments, the IXplore Standard microscope system is easy to operate and capable of excellent, publication-quality images. A range of encoded unit options makes it easy to get accurate and reproducible results.

IXplore Live

Live Cell Imaging Microscope System

Designed for precise live cell imaging, the IXplore Live microscope system helps reduce photobleaching and enhance cell viability for physiological experiments.

IXplore Spin

Confocal Imaging Microscope System

The IXplore Spin microscope system's advanced spinning disk unit contributes to a large field of view, fast 3D confocal image acquisition, and prolonged cell viability in time-lapse experiments.

IXplore Pro

Automated Microscope System

Designed for automated multidimensional observation and easy experiment setup, the IXplore Pro microscope system facilitates effortless data collection and seamless acquisition of panoramic, multichannel images.

IXplore TIRF

TIRF Imaging Microscope System

Designed for membrane dynamics, single molecule detection, and colocalization experiments, the IXplore TIRF microscope system offers simultaneous multicolor TIRF imaging for up to 4 colors with high stability.

IXplore SpinSR

Super Resolution Microscope System

Designed for fast 3D super resolution imaging and prolonged cell viability in time-lapse experiments, the IXplore SpinSR microscope system offers XY resolution down to 120 nm without the need for dedicated labeling procedures.

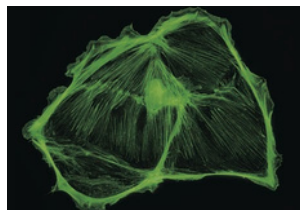
Inverted Imaging Platform

Specifications

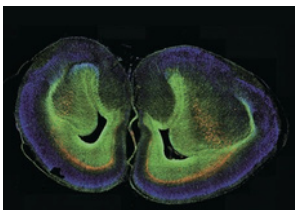
		IXplore Standard	IXplore Pro	IXplore Live	IXplore TIRF	IXplore Spin	IXplore SpinSR
Microscope frame	IX73	✓					
	IX83		✓	✓	✓	✓	✓
Transmitted Köhler illumination	12 V 100 W halogen (U-LH100L)	✓					
	High color rendering LED (IX3-LHLEDC)		✓	✓	✓	✓	✓
Stage	Mechanical stage with right handle (IX3-SVR)	✓					
	Ultrasonic scanning stage (IX3-SSU)		✓	✓	✓	✓	✓
Condenser	Long working distance universal (IX3-LWUCD)	✓					
	Motorized long working distance universal (IX3-LWUCDA)		✓	✓	✓	✓	✓
Fluorescence illuminator	L-shaped fluorescence illuminator with fly-eye lens (IX3-RFALFE)	✓	✓	✓	✓	✓	✓
Fluorescence mirror turret	Coded fluorescence mirror turret (IX3-RFACS)	✓					
	Motorized fluorescence mirror turret (IX3-RFACA)		✓	✓	✓	✓	✓
Fluorescence mirror unit	UIS2 mirror units	✓	✓	✓	✓	✓	✓
Fluorescence light source	LED and LDP light source (U-LGPS)	✓	✓	✓	✓	✓	✓
Imaging software	cellSens Standard	✓					
	cellSens Dimension		✓	✓	✓	✓	✓
TIRF illuminator	cellTIRF				✓		
Confocal scanner	Spinning disk confocal scanner					✓	✓
Super resolution processing	Olympus super-resolution (OSR) filter						✓
	TruFocus system (IX3-ZDC2)			✓	✓	✓	✓
Accessories	Remote correction collar controller (IX3-RCC)			✓	✓	✓	✓
	Real-time controller (U-RTC/U-RTCe)			✓	✓	✓	✓
	Incubation housing			✓	✓	✓	✓

Observation Images

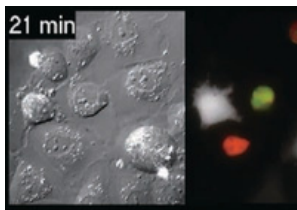
IXplore Standard



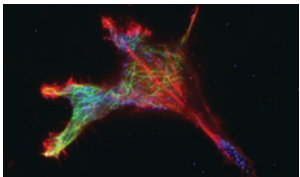
IXplore Pro



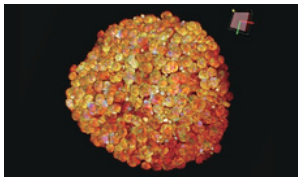
IXplore Live



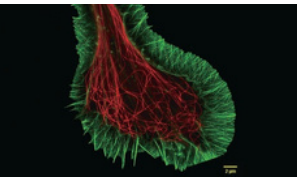
IXplore TIRF



IXplore Spin*



IXplore SpinSR*



EVIDENT
OLYMPUS

LABEX
FLUOCCA
IMARIS

GREINER
BIOSCIENCE

APPLIED
SPECTRAL
IMAGING

SAKURA
FINETEK

ZTEC
CRAFTK

TEKSQRAY
FEATHER

DYNAMICA
FROILABO

FASTER
ARCTIKO
KUHNER
HETTICH

HBK
CAMECA
IX CAMERAS

THERMO EM
CRYSTAL

IX Applications for ICSI/IMSI

Optimized for a Smoother ICSI Workflow

IX73 Microscope

Motorized Observation Method Switching

- Quickly change observation methods by pressing a button on the hand switch
- Spindle observation is standard with a semi-motorized system; IMSI units are available



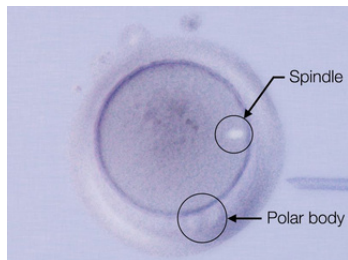
IX83 Microscope

Fully Motorized and Highly Advanced

- Standard configuration supports IMSI and spindle observation
- Quickly change observation methods and magnification using the controller
- Focus position and brightness automatically adjust when you change magnifications
- Easily operate the focus and switch between camera and visual observation using the remote control, reducing sample vibration



Applied Technologies



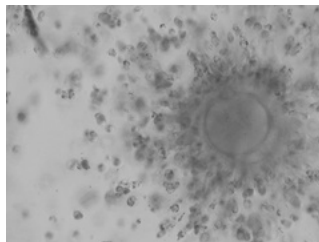
Easy Oocyte Condition Checks through Spindle Visualization

Olympus' observation method enables you to easily view the metaphase II (MII) spindle through the microscope oculars



Relief Contrast

Olympus' relief contrast enables 3D imaging of oocytes so you can check the condition of the zona pellucida



Polarization

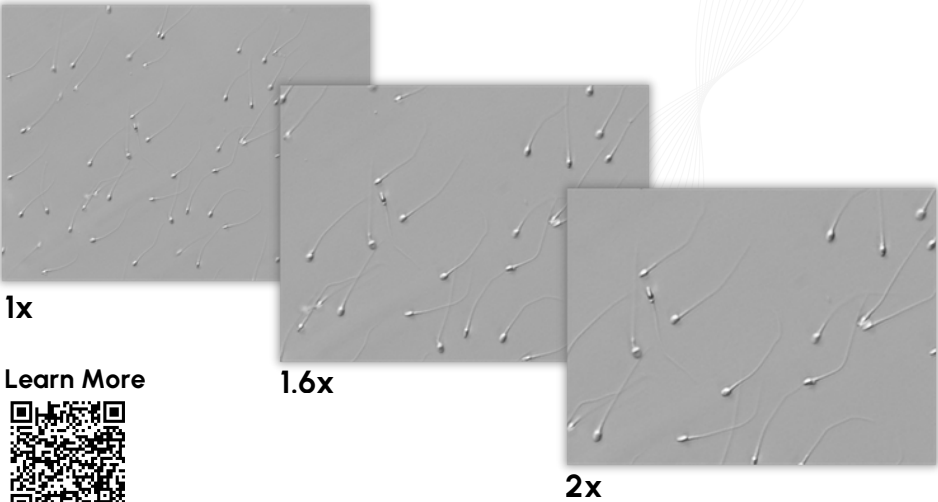
Observe the birefringence of spindles using polarized light

		IX73	IX83
Microscope frame	Optical system	UIS2 optical system	
	Revolving nosepiece	- Motorized sextuple revolving nosepiece (DIC slider attachable), simple waterproof structure - Coded sextuple revolving nosepiece (DIC slider attachable), simple waterproof structure	Motorized sextuple revolving nosepiece (DIC slider attachable), simple waterproof structure
	Focus	Stroke: 10 mm	Stroke: 10.5 mm Minimum increment: 0.01 μ m, Maximum nosepiece movement speed: 3 mm/s
	Intermediate port	2 ports (IX73P2F), 1 port (IX73PIF)	1 port (IX83PIZF)
	Light path selection	Manual 0.100/50.50/100.0 (Left side port: BI port)	Motorized 0.100/50.50/100.0 (Left side port: BI port)
Transmitted light illuminator		Pillar tilt mechanism (30° inclination angle, with vibration reducing mechanism), Condenser holder (with with 88 mm stroke, refocusing mechanism), Field iris diaphragm adjustable, 4 filter holders Light source: 12 V, 100 W halogen bulb (pre-centered) - High color reproductive LED light source	
Observation tube	Widefield (FN 22)	- Widefield tilting binocular - Widefield binocular	
Stage	Mechanical stage with left handle	Stage stroke: X: 50 mm x Y: 43 mm, including stage insert plate (ø25 mm), stage insert plate exchangeable (Ø110 mm)	
	Cross stage with right handle	Stage stroke: X: 50 mm x Y: 50 mm, including stage insert plate (ø25 mm), stage insert plate exchangeable (Ø110 mm)	
	Mechanical stage with right handle	Stage stroke: X: 114 mm x Y: 75 mm, stage position locking function	
Condenser	Motorized middle long working distance condenser	NA 0.5, W.D. 45 mm, 4 positions for optical devices (for ø50 mm, relief contrast optical devices rotatable)	
	Mid long working distance relief contrast	NA 0.5, W.D. 45 mm, 4 positions for optical devices (for ø50 mm, relief contrast optical devices rotatable)	
Operating environment	- Indoor use - Ambient temperature: 5° to 40°C (41° to 104°F) - Maximum relative humidity: 80% for temperatures up to 31°C (88°F), decreasing linearly through 70% at 34°C (93°F), 60% at 37°C (99°F), to 50% relative humidity at 40°C (104°F) - Supply voltage fluctuations: Not to exceed ±10% of the normal voltage		

Morphological Evaluation of Sperm at High Magnification

Improve the sperm selection process:

- Smooth zooming from 1x, 1.6x, to 2x using the adjustment lever.
- Combine the zoom with the 40X objective lens to observe the morphology and internal structure of the sperm head in detail.



Learn More

